

MongoDB Practical: Restaurants & Indexing

Collection creation, queries, single-field index, compound index, performance comparison, and optimized top-5 query.

1. Create Collection, Insert 10 Documents & Find Italian Restaurants

MongoDB Commands

```
use restaurantDB

db.createCollection("restaurants")

db.restaurants.insertMany([
  { name: "La Piazza", cuisine: "Italian", location: "Surat", rating: 4.5 },
  { name: "Pasta Palace", cuisine: "Italian", location: "Ahmedabad", rating: 4.3 },
  { name: "Dragon House", cuisine: "Chinese", location: "Ahmedabad", rating: 4.2 },
  { name: "Spice Garden", cuisine: "Indian", location: "Surat", rating: 4.6 },
  { name: "Roma Kitchen", cuisine: "Italian", location: "Vadodara", rating: 4.7 },
  { name: "Pizza Hub", cuisine: "Pizza", location: "Surat", rating: 4.8 },
  { name: "Golden Wok", cuisine: "Chinese", location: "Ahmedabad", rating: 4.4 },
  { name: "Bella Italia", cuisine: "Italian", location: "Surat", rating: 4.9 },
  { name: "Tandoori Tales", cuisine: "Indian", location: "Ahmedabad", rating: 4.1 },
  { name: "Pasta Corner", cuisine: "Italian", location: "Rajkot", rating: 4.4 }
])

db.restaurants.find({ cuisine: "Italian" })
```

Output

La Piazza	Italian	Surat	4.5
Pasta Palace	Italian	Ahmedabad	4.3
Roma Kitchen	Italian	Vadodara	4.7
Bella Italia	Italian	Surat	4.9
Pasta Corner	Italian	Rajkot	4.4

2. Single-Field Index on cuisine & Execution-Time Comparison

Before Index

```
db.restaurants.find({ cuisine: "Italian" }).explain("executionStats")

Sample output:
executionTimeMillis: 1
winningPlan: COLLSCAN
```

Create Index

```
db.restaurants.createIndex({ cuisine: 1 })

Output:
"cuisine_1"
```

After Index

```
db.restaurants.find({ cuisine: "Italian" }).explain("executionStats")

Sample output:
executionTimeMillis: 0
winningPlan: IXSCAN
indexName: "cuisine_1"
```

Performance Comparison

Condition	Sample Time	Query Plan
Before index	1 ms	COLLSCAN
After index	0 ms	IXSCAN

Note: Execution time varies by MongoDB version, hardware, cache state, and collection size. With only 10 documents, the difference may be negligible.

3. Compound Index on cuisine and location

```
db.restaurants.createIndex({ cuisine: 1, location: 1 })
```

Output:
"cuisine_1_location_1"

```
db.restaurants.find({
  cuisine: "Chinese",
  location: "Ahmedabad"
})
```

Output:
Dragon House
Golden Wok

Check Query Planner

```
db.restaurants.find({
  cuisine: "Chinese",
  location: "Ahmedabad"
}).explain("executionStats")
```

Relevant output:
stage: "IXSCAN"
indexName: "cuisine_1_location_1"

The IXSCAN stage and the compound index name indicate that MongoDB is using the compound index.

4. Drop the cuisine Index & Observation

```
db.restaurants.dropIndex("cuisine_1")
```

Output:
{ ok: 1 }

```
db.restaurants.find({ cuisine: "Italian" }).explain("executionStats")
```

Observation

After dropping the single-field cuisine index, MongoDB may scan more documents when filtering only by cuisine. The difference is usually small for 10 documents but can become significant as the collection grows. The compound cuisine+location index may still support queries that use cuisine because cuisine is its leading field.

5. Top 5 Highest-Rated Pizza Restaurants in Surat

Recommended index

```
db.restaurants.createIndex({
  cuisine: 1,
  location: 1,
  rating: -1
})
```

Output:
"cuisine_1_location_1_rating_-1"

Query

```
db.restaurants.find({
  cuisine: "Pizza",
  location: "Surat"
})
.sort({ rating: -1 })
.limit(5)
```

Check Index Usage

```
db.restaurants.find({
  cuisine: "Pizza",
  location: "Surat"
})
.sort({ rating: -1 })
.limit(5)
.explain("executionStats")
```

Relevant output:
stage: "IXSCAN"
indexName: "cuisine_1_location_1_rating_-1"

Explanation: Indexing helps MongoDB quickly locate Pizza restaurants in Surat and retrieve them in rating order, reducing document scanning and sorting work.

Final Practical Summary

Task	Result
Italian search	Finds 5 Italian restaurants
Single-field index	cuisine_1
Compound index	cuisine_1_location_1
Index verification	IXSCAN / indexName
Dropped index	cuisine_1 removed
Top-5 optimization	cuisine + location + rating index